

A. Parsa's Humongous Tree

time limit per test: 1 second
memory limit per test: 256 megabytes

Parsa has a humongous tree on n vertices.

On each vertex v he has written two integers l_v and r_v .

To make Parsa's tree look even more majestic, Nima wants to assign a number a_v ($l_v \leq a_v \leq r_v$) to each vertex v such that the beauty of Parsa's tree is maximized.

Nima's sense of the beauty is rather bizarre. He defines the beauty of the tree as the sum of $|a_u - a_v|$ over all edges (u, v) of the tree.

Since Parsa's tree is too large, Nima can't maximize its beauty on his own. Your task is to find the **maximum** possible beauty for Parsa's tree.

Input

The first line contains an integer t ($1 \leq t \leq 250$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains a single integer n ($2 \leq n \leq 10^5$) — the number of vertices in Parsa's tree.

The i -th of the following n lines contains two integers l_i and r_i ($1 \leq l_i \leq r_i \leq 10^9$).

Each of the next $n - 1$ lines contains two integers u and v ($1 \leq u, v \leq n, u \neq v$) meaning that there is an edge between the vertices u and v in Parsa's tree.

It is guaranteed that the given graph is a tree.

It is guaranteed that the sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Output

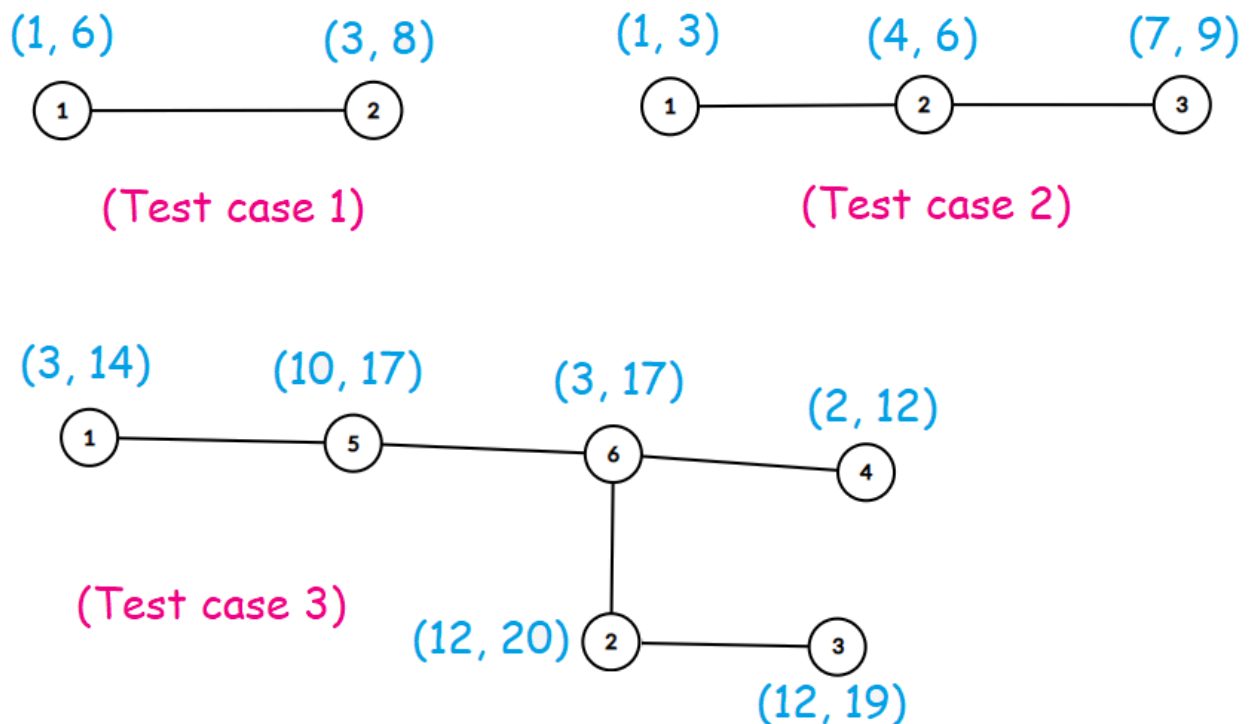
For each test case print the **maximum** possible beauty for Parsa's tree.

Example

input	output
3	7
2	8
1 6	62
3 8	
1 2	
3	
1 3	
4 6	
7 9	
1 2	
2 3	
6	
3 14	
12 20	
12 19	
2 12	
10 17	
3 17	
3 2	
6 5	
1 5	
2 6	
4 6	

Note

The trees in the example:



In the first test case, one possible assignment is $a = \{1, 8\}$ which results in $|1 - 8| = 7$.

In the second test case, one of the possible assignments is $a = \{1, 5, 9\}$ which results in a beauty of $|1 - 5| + |5 - 9| = 8$